After some weeks there came a response from the editor at the University of Chicago: The manuscript had been sent to the American physicist Carl Eckart for refereeing. In the manuscript Chandrasekhar stated, without explanation, the result of his relativistic and quantum mechanical calculation, that the resistance to compression is ⅓ at ultra-high densities. This ⅓ resistance was essential to the limit on how heavy a white dwarf can be. If the resistance were larger than ⅓, then white dwarfs could be as heavy as they wished—and Eckart thought it should be larger. Chandrasekhar fired off a reply containing a mathematical derivation of the ⅓ resistance; Eckart, reading the details, conceded that Chandrasekhar was right and approved his paper for publication. Finally, a full year after Chandrasekhar had written it, his paper got published.[4]

The response of the astronomical community was deafening silence. Nobody seemed interested. So Chandrasekhar, wanting to complete his Ph.D. degree, turned to other, more acceptable research.

Three years later, with his Ph.D. finished, Chandrasekhar visited Russia to exchange research ideas with Soviet scientists. In Leningrad a young Armenian astrophysicist, Viktor Amazapovich Ambartsumian, told Chandrasekhar that the world's astronomers would not believe his strange limit on the masses of white dwarfs unless he computed, from the laws of physics, the masses of a representative sample of white dwarfs and demonstrated explicitly that they were all below his claimed limit. It was not enough, Ambartsumian asserted, that Chandrasekhar had analyzed white dwarfs with rather low densities and resistances of ⅗, and white dwarfs with extremely high densities and resistances of ⅓; he needed also to analyze a goodly sample of white dwarfs with densities in between and show that they, too, always have masses below 1.4 Suns. Upon returning to Cambridge, Chandrasekhar took up Ambartsumian's challenge.

One foundation that Chandrasekhar would need was the *equation of state* of white-dwarf matter over the entire range of densities, running from low to extremely high. (By the "state" of the matter, physicists mean the matter's density and pressure—or equivalently its density and its resistance to compression, since from the resistance and the density one can compute